AASTHA AMISH KHATRI

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EDUCATION

Bachelor of Technology (B.Tech), Computer Science & Engineering

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar (CGPA: 8.1 upto 5th semester)

Expected Graduation: 2027

Relevant Coursework: Software Engineering, Database Management Systems, Computer Organisation and Architecture, Machine Learning, Artificial Intelligence, Cloud Computing, Computer Intelligence, Computer Networks, Data Structures and Algorithms

SKILLS

Languages and Databases: Python, Java, C, MySQL

Frameworks/Libraries: PyTorch, Tensorflow, NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV, Gymnasium

Web: HTML, CSS, JavaScript

Tools: Git, VS Code, Kaggle, Jupyter, AWS (S3, EC2, IAM, Lambda, VPC)

WORK EXPERIENCE

Nebula Space Organisation | Bhubaneswar, India Dec 2023 - Feb 2025

Research & Development Intern:

- Automated data extraction pipelines in Python processing structured data across multiple spreadsheets, reducing manual effort for internal reporting
- Conducted systematic review of 50+ technical papers on satellite subsystems, synthesizing material selection criteria for specific engineering use cases

PROJECTS

Adaptive Image Compression using CNN-based Importance Maps Python, OpenCV, NumPy, PyTorch, DCT

- Developed adaptive image compression using CNN importance maps with a JPEG (DCT) pipeline and block-wise adaptive quantization
- Outperformed standard JPEG by +0.14 dB PSNR (34.55 vs 34.41 dB) at comparable bitrate, validated across 137 held-out test images at Q=40
- Achieved 9x compression ratio (192 KB → 21 KB) while maintaining higher perceptual quality than standard JPEG

Deep Reinforcement Learning – Atari Ms. Pacman (DQN) Python, PyTorch, Gymnasium (Atari), NumPy, OpenCV

- Built a DQN agent in PyTorch on raw pixel input (64×64) with 4-frame stacking — no handcrafted features
- Achieved ~54% reward improvement; evaluation scores 500–700, best episode 1,400+ points
- Ablation study on frame stacking quantified its impact on motion and directional inference

Comparative Analysis of Deep Q-Network Variants for LunarLander-v3 Python, PyTorch, Gymnasium (Box2D), NumPy

- Compared four DQN variants in PyTorch across 1,500 episodes on LunarLander-v3
- Vanilla DQN achieved Avg100: 233, converging ~400 episodes faster than all variants
- Dueling DQN showed critical instability at ep. 850–900, highlighting that architectural complexity doesn't guarantee improvement on well-shaped reward environments

Advanced MobileNet-SSD Based Object Recognition System for Visually Impaired Assistance Python, TensorFlow, OpenCV, MobileNet-SSD, Computer Vision, Text-to-Speech

- Built a real-time object detection system using MobileNet-SSD with ~71% accuracy in live environments
- Optimized performance using SSIM-based frame skipping, reducing redundant computation
- Integrated text-to-speech feedback with duplicate filtering for improved user experience

Continuous Control for Simulated Driving using PPO and SAC (Feb 2026 - Present) Python, PyTorch, Gymnasium, NumPy

- Developing and training Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC) agents for continuous-control tasks in a simulated environment
- Exploring training stability, convergence behavior, and reward optimization across policy-gradient methods

ResarchFlow - Hybrid Research Assistant (April 2026 - Present) Python, FastAPI, LangChain, React, AWS, Gemini API

- A full-stack RAG assistant with FastAPI, LangChain, and ChromaDB supporting hybrid document + web search routing
- Integrating Gemini API for citation-aware answers, PostgreSQL for session history, and multi-paper comparison and deploying on AWS EC2 and Vercel

COURSES AND CERTIFICATIONS

- AWS Academy Graduate – Cloud Foundations (AWS Academy) – Verified badge available on Credly
- Artificial Intelligence: Agentic AI, Generative AI, Reinforcement Learning – Udemy
- The AI Engineer Course 2026: Complete AI Engineer Bootcamp – Udemy (in progress)
- CS50's Introduction to Artificial Intelligence with Python (HarvardX) - edX
- Data Science: Machine Learning (HarvardX) - edX
- The Complete Full-Stack Web Development Bootcamp – Udemy (in progress)